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(19) **United States**(12) **Patent Application Publication**
LEE(10) **Pub. No.: US 2025/0275792 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **SPACING-ADJUSTABLE ORTHOPEDIC APPARATUS***A61B 17/68* (2006.01)*A61B 17/86* (2006.01)(71) Applicant: **KOREA UNIVERSITY RESEARCH AND BUSINESS FOUNDATION,**
Seoul (KR)(52) **U.S. Cl.**CPC *A61B 17/8009* (2013.01); *A61B 17/8071* (2013.01); *A61B 17/8605* (2013.01); *A61B 17/8625* (2013.01); *A61B 17/8685* (2013.01); *A61B 17/8695* (2013.01); *A61B 2017/00398* (2013.01); *A61B 2017/00477* (2013.01); *A61B 2017/00862* (2013.01); *A61B 2017/681* (2013.01)(72) Inventor: **Eui Seok LEE, Seoul (KR)**(21) Appl. No.: **18/262,189**(22) PCT Filed: **Jan. 18, 2022**(86) PCT No.: **PCT/KR2022/000867**

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The present invention relates to a spacing-adjustable bone orthopedic apparatus. The apparatus may include a plate including a main body and a through portion having a predetermined length and formed to pass through the main body and a locking screw which is inserted into the through portion, which includes a screw body and a screw head which are capable of being coupled to each other, and in which outer circumferential surface saw teeth are formed on an outer circumferential surface of the screw head.

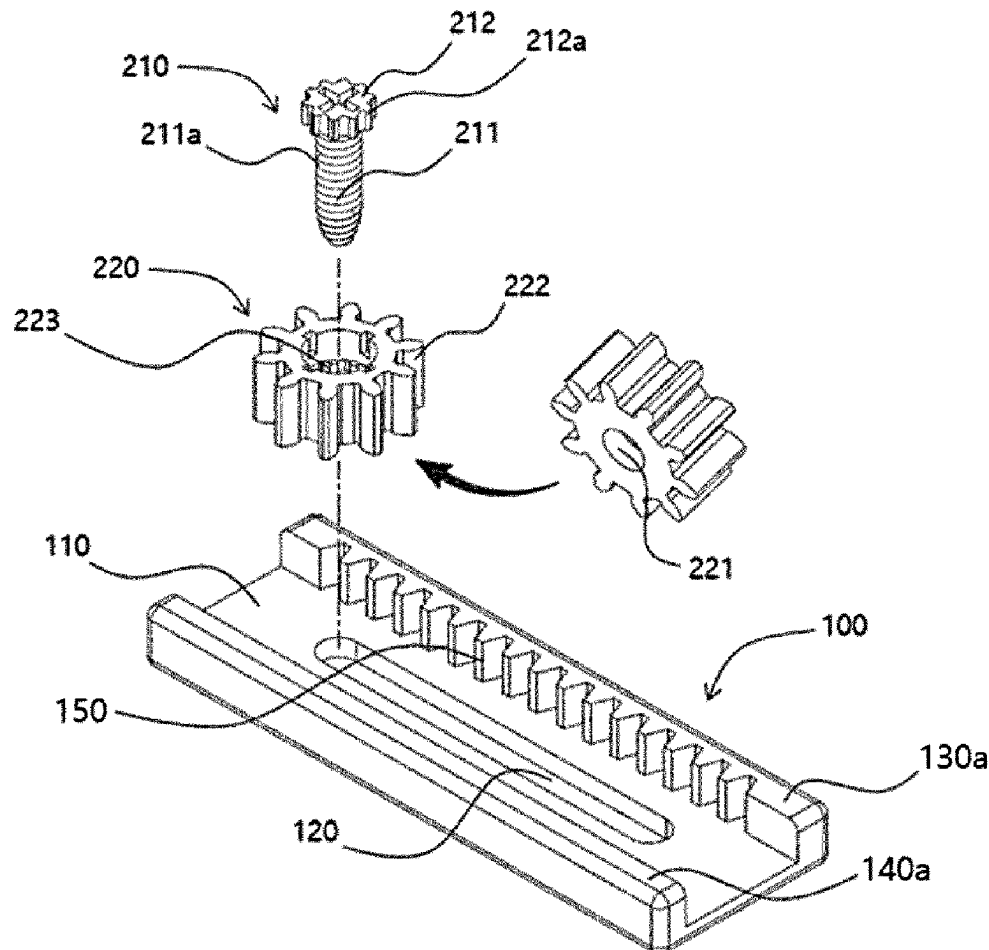


FIG. 1

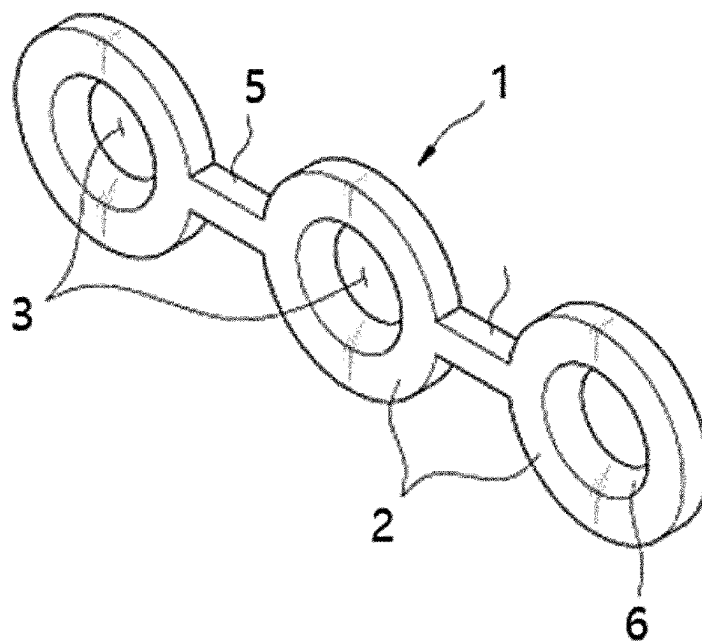


FIG. 2

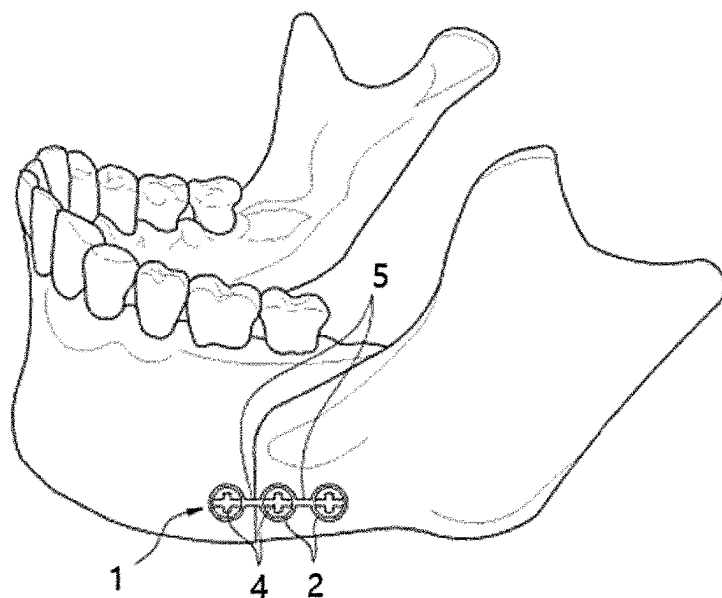


FIG. 3

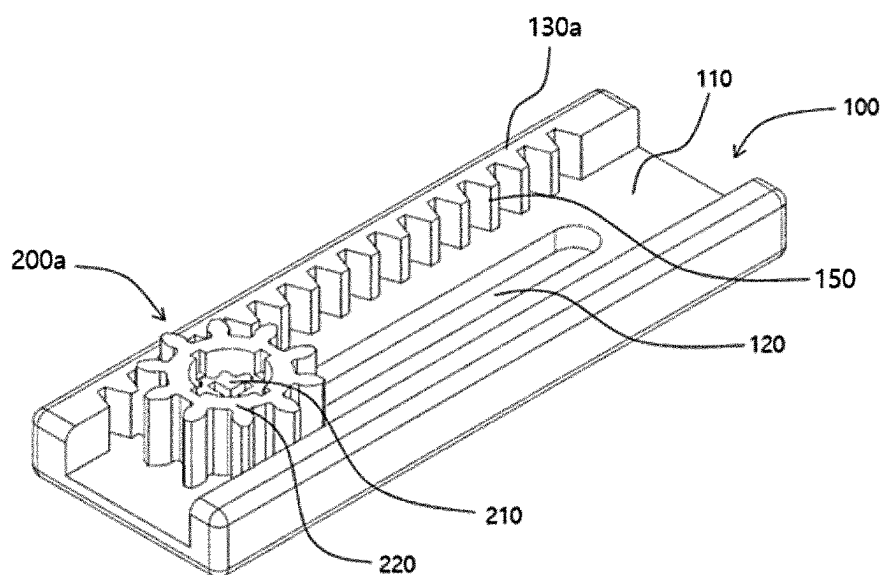


FIG. 4

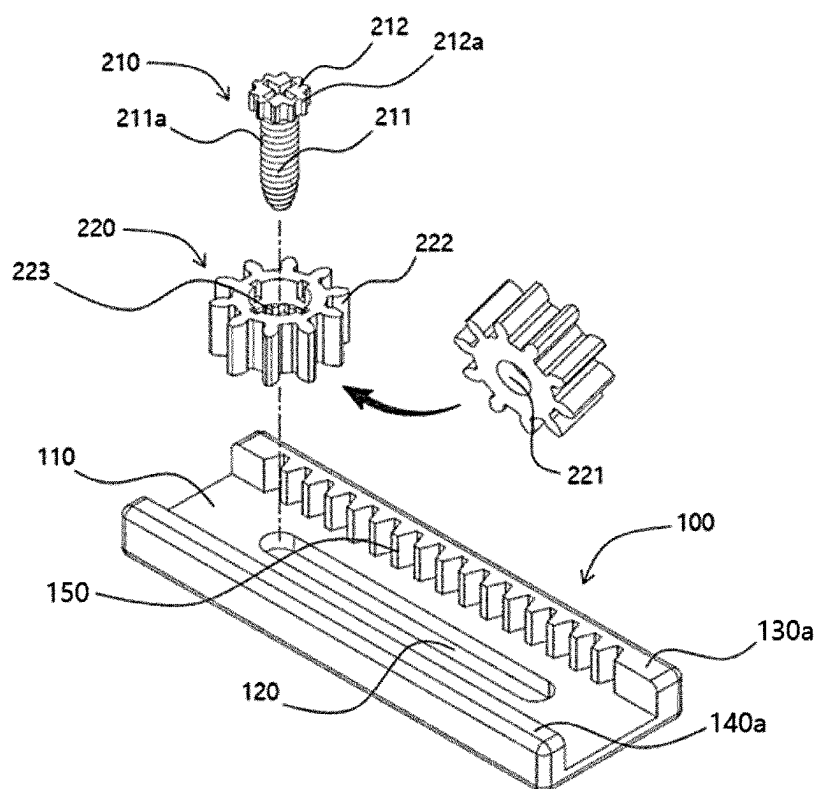
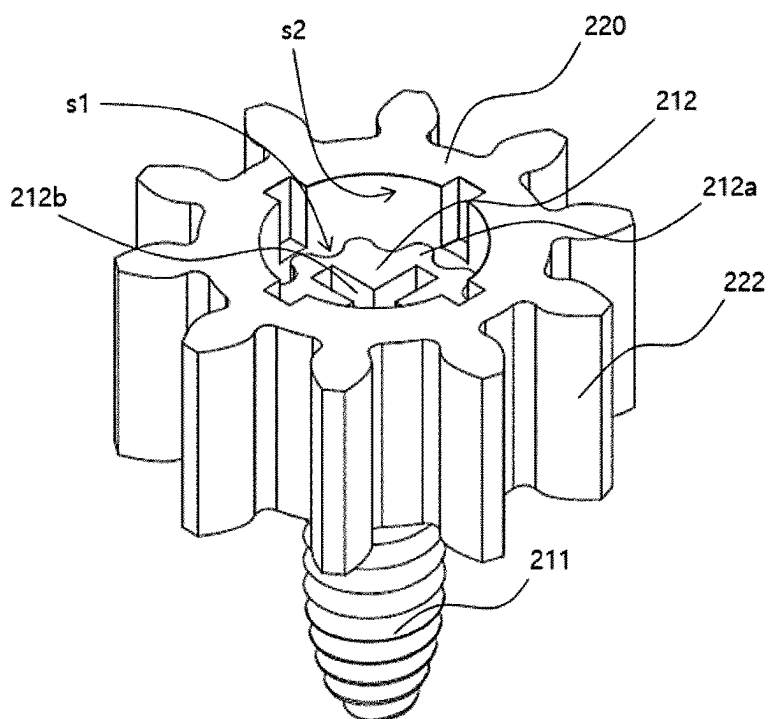


FIG. 5



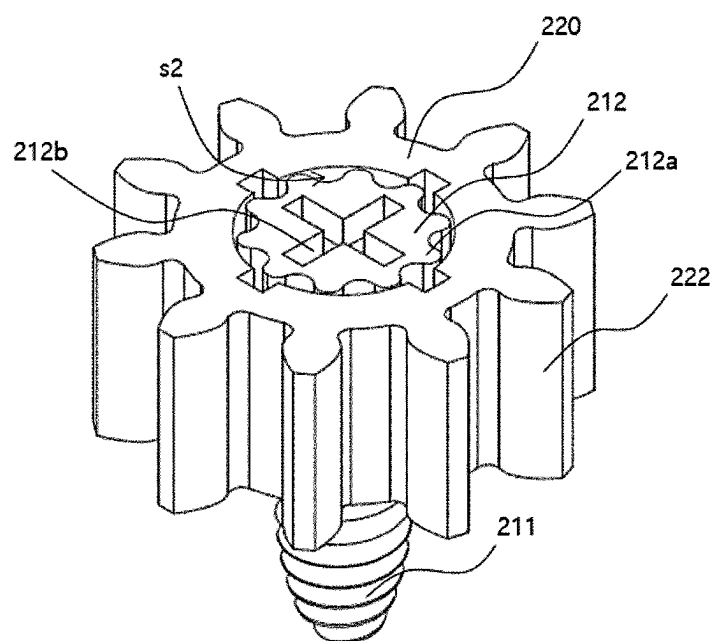


FIG. 7

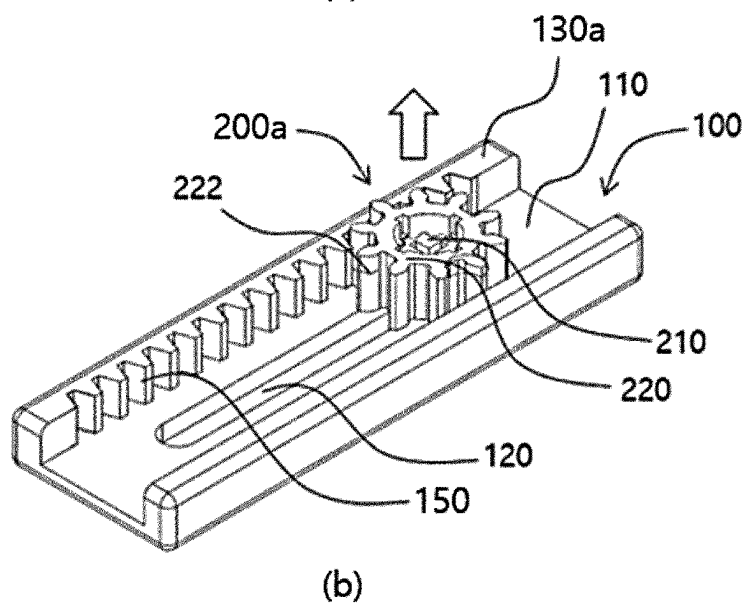
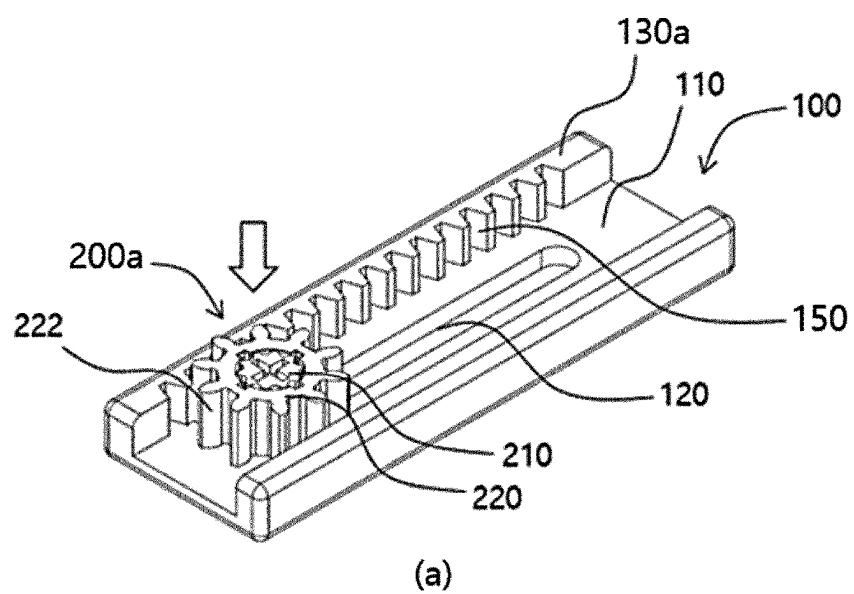


FIG. 8

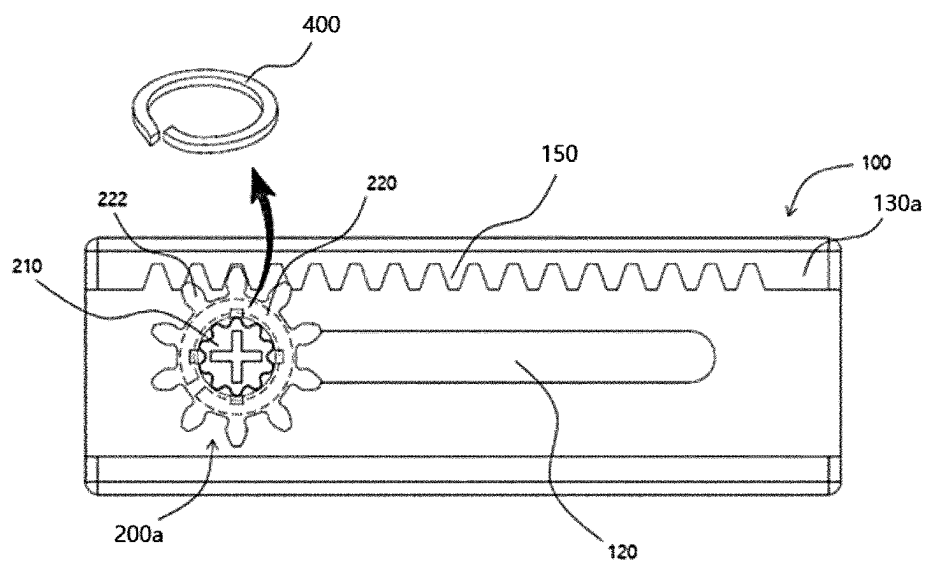


FIG. 9

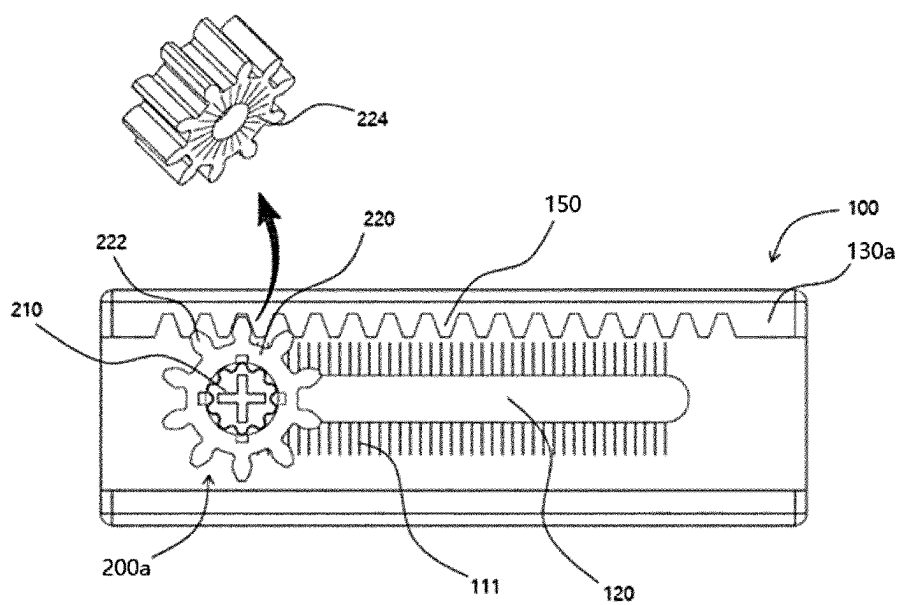


FIG. 10

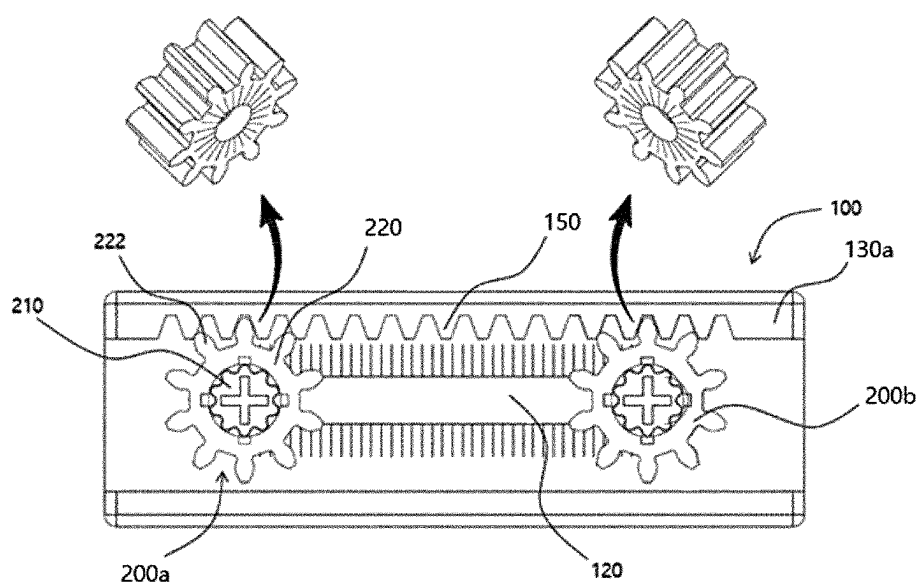


FIG. 11

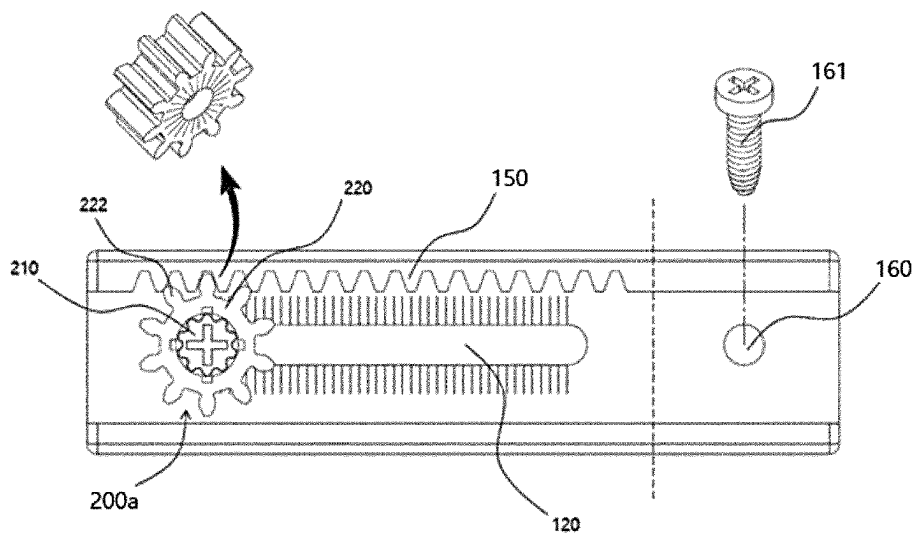


FIG. 12

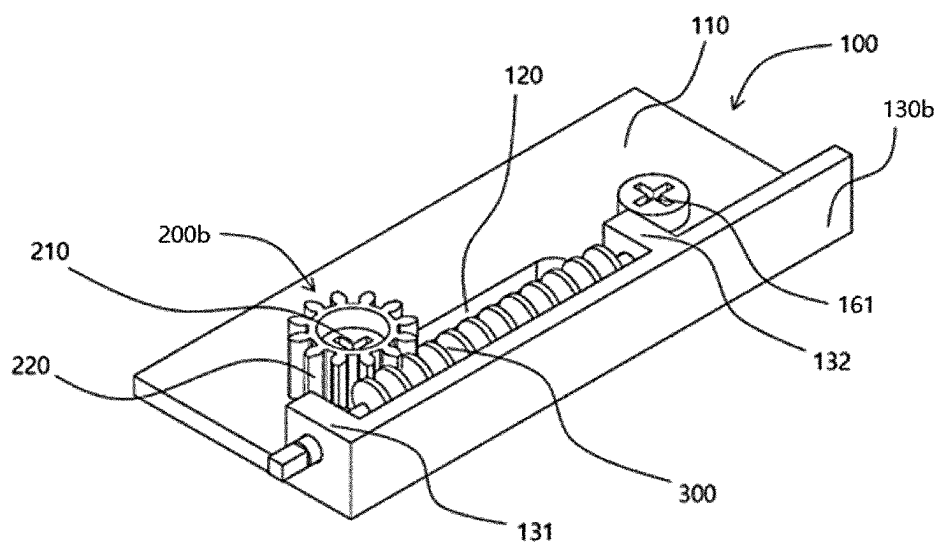


FIG. 13

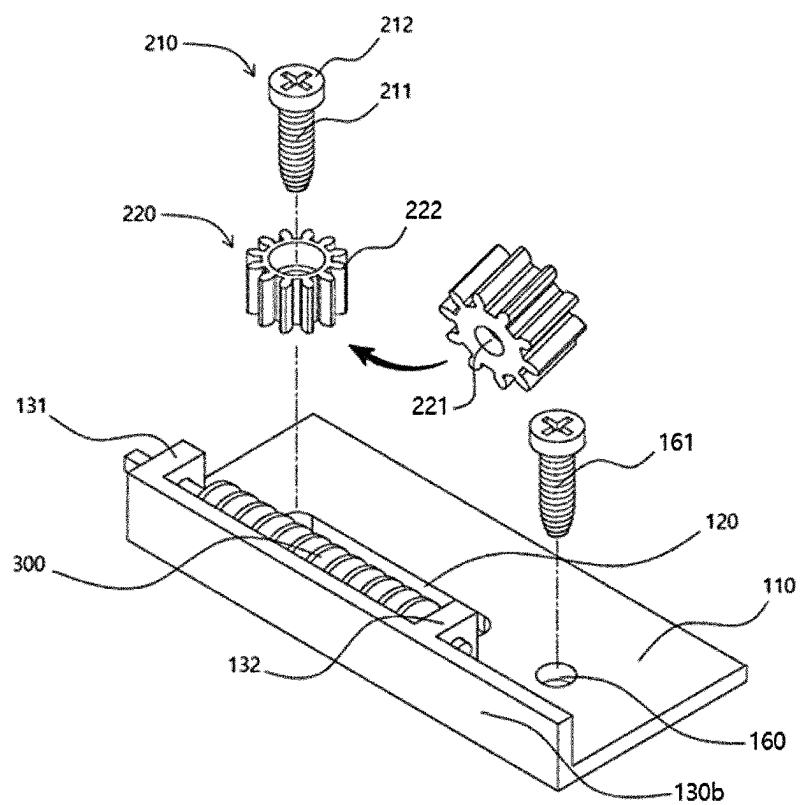


FIG. 14

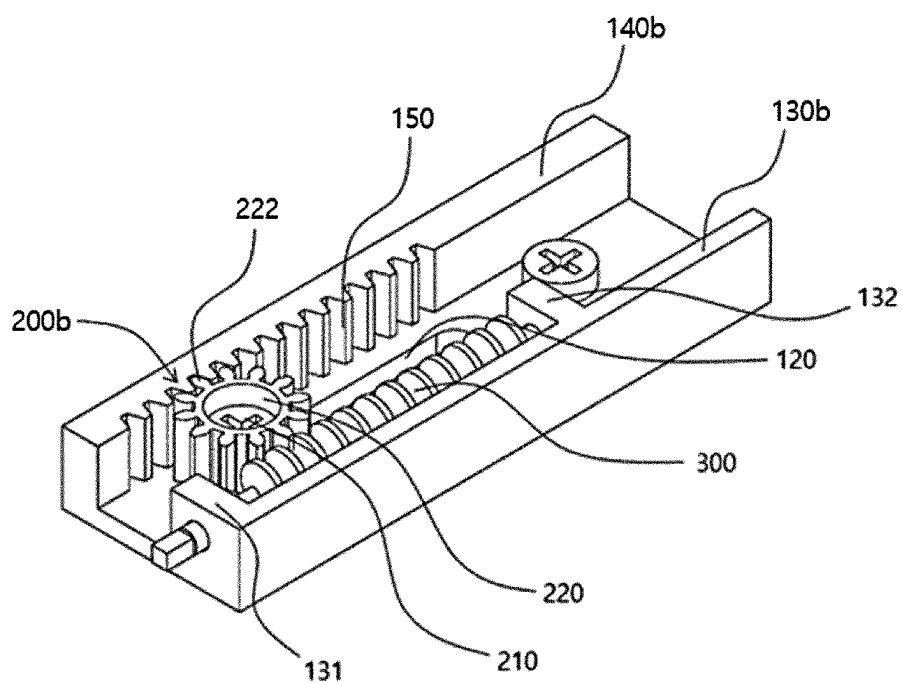
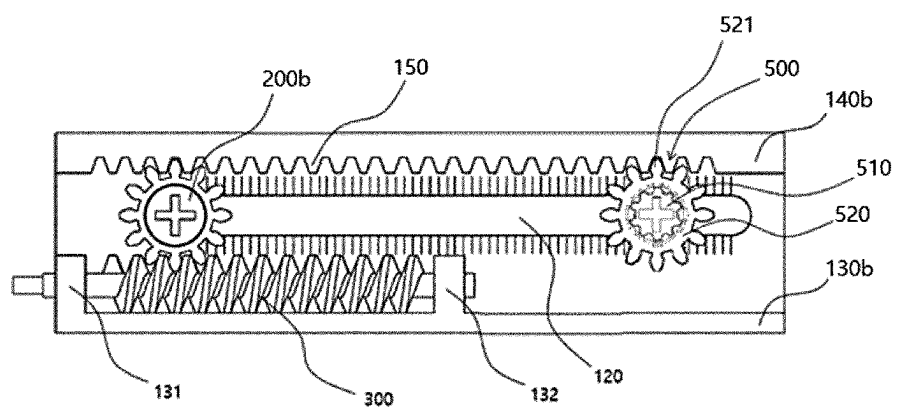


FIG. 15



SPACING-ADJUSTABLE ORTHOPEDIC APPARATUS

TECHNICAL FIELD

[0001] The present invention relates to a spacing-adjustable bone orthopedic apparatus for adjusting the relative positions of two bone fragments during bone orthopedic surgery.

BACKGROUND ART

[0002] Generally, a mandibular fixation plate is an auxiliary medical device used, after a part of a jawbone is removed or incised when performing facial cosmetic surgery for a chin for correcting a protruding chin or oral cancer surgery or after an abnormal portion is found in a bone and the abnormal portion is removed, to maintain a shape of the remaining portion or transplant and fix it with new bone.

[0003] Hereinafter, a conventional method of applying a mandibular fixation plate to the mandible of a face will be described with reference to FIGS. 1 and 2 as follows.

[0004] FIG. 1 is a perspective view illustrating a conventional fixation plate for mandibular surgery, and FIG. 2 is a perspective view illustrating the fixation plate for mandibular surgery of FIG. 1 attached to the mandible.

[0005] Referring to FIGS. 1 and 2, in a state in which a part of the mandible is removed or incised, the fixation plate for mandibular surgery is mounted on a front surface of the mandible, and a conventional plate body 1 for mandibular surgery is formed of a material with a predetermined strength or more. A plurality of ring-shaped fixing parts 2 in which through portions 3 are formed at equal intervals are formed in the plate body 1. Screws 4 are inserted into the mandible through the through portions 3, and thus the plate body 1 is fastened to the mandible. The fixing parts 2 are mutually connected through connecting parts 5, and tapered surfaces 6 for allowing head portions of the fastened screws 4 to be seated are formed on inner circumferential surfaces of the fixing parts 2.

[0006] An operator inserts the screws 4 into the through portions 3 formed in the fixing parts 2 and then fastens the screws 4 to the mandible to complete a joining surgical procedure of the incised mandible.

[0007] Meanwhile, during the joining surgical procedure of the mandible, since a patient is under anesthesia, and the patient has not sensation, consciousness, and muscle activity, in a state in which a head is fixed using a tool, the fixation plate is fixed while the mandible is positioned according to the operator's experience.

[0008] However, as a position of the fixation plate slightly is changed in upward, downward, left, and right directions for each patient after the surgical procedure is performed due to a different growth rate and a growth direction of the mandible, a tooth structure, and chewing habits, the position of the conventional mandibular fixation plate needs to be finely adjusted. However, when the position of the conventional fixation plate is adjusted, a fastening state of the screws 4 fastened to the through portions 3 of the fixing parts 2 formed in the plate body 1 is released and then the screws 4 should be fixed to new positions, and thus there is a problem that it is actually very difficult to adjust the position of the fixation plate.

RELATED ART

Patent Document

[0009] (Patent Document 1) Korean Registered Patent No. 10-1210089

Technical Problem

[0010] The present invention is directed to providing a spacing-adjustable bone orthopedic apparatus capable of finely adjusting the relative positions of two bone fragments after surgery.

Technical Solution

[0011] One aspect of the present invention provides a spacing-adjustable bone orthopedic apparatus, and the apparatus includes a plate including a main body and a through portion having a predetermined length and formed to pass through the main body and a locking screw which is inserted into the through portion, which includes a screw body and a screw head which are capable of being coupled to each other, and in which outer circumferential surface saw teeth are formed on an outer circumferential surface of the screw head.

[0012] According to a first embodiment of the present invention, the plate may include a sidewall formed at one side of the main body, a plurality of plate saw teeth may be formed on the sidewall in a longitudinal direction of the sidewall to extend toward the through portion, the screw body may include a body portion in which a screw thread is formed on a lower outer circumferential surface of the screw body and a head portion in which locking saw teeth are formed on an upper outer circumferential surface of the screw body, the outer circumferential surface saw teeth of the screw head may correspond to a shape of the plate saw teeth, and inner circumferential surface saw teeth corresponding to a shape of the locking saw teeth of the head portion of the screw body may be formed on a part of an inner circumferential surface of the screw head.

[0013] According to the first embodiment of the present invention, the screw head may be disposed on the through portion and supported by an upper surface of the main body of the plate.

[0014] According to the first embodiment of the present invention, the inner circumferential surface of the screw head may include a locking region in which the inner circumferential surface saw teeth are formed and a locking release region other than the locking region, wherein the locking region may be formed to extend inside the inner circumferential surface of the screw head and has a predetermined height.

[0015] According to the first embodiment of the present invention, it may be designed that a locking protrusion protruding inward is formed on a portion adjacent to a lowermost side of the inner circumferential surface of the screw head and a lower side surface of the head portion of the screw body is caught by the locking protrusion.

[0016] According to the first embodiment of the present invention, the spacing-adjustable bone orthopedic apparatus may further include an elastic member which is disposed between a lower surface of the screw head and the main body and of which one side is inclined toward the lower surface of the screw head.

[0017] According to the first embodiment of the present invention, the elastic member may be a spring washer.

[0018] According to the first embodiment of the present invention, the spacing-adjustable bone orthopedic apparatus may further include head ratchet saw teeth formed on a lower side surface of the screw head and body ratchet saw teeth formed on an upper side surface of the main body and corresponding to a shape of the head ratchet saw teeth.

[0019] According to the first embodiment of the present invention, the locking screw may include a first locking screw and a second locking screw disposed in the through portion to be spaced apart from the first locking screw in a longitudinal direction of the through portion.

[0020] According to the first embodiment of the present invention, a screw fixing hole spaced apart from the through portion may be formed in the plate.

[0021] According to a second embodiment of the present invention, the plate may include a first support part and a second support part which are formed to protrude from a bottom surface at one side of the main body and are spaced apart from each other, wherein the spacing-adjustable bone orthopedic apparatus may include a worm which is supported by the first support part and the second support part and on which a screw thread that engages with the outer circumferential surface saw teeth of the locking screw is formed, and the locking screw may be used as a worm wheel.

[0022] According to the second embodiment of the present invention, one end portion of the worm may pass through and may be supported by the first support part, and the other end portion of the worm may pass through and may be supported by the second support part.

[0023] According to the second embodiment of the present invention, the spacing-adjustable bone orthopedic apparatus may further include a motor which is connected to the worm and rotates the worm in one direction or the other direction.

[0024] According to the second embodiment of the present invention, the plate may further include a sidewall formed at the other side of the main body, and a plurality of plate saw teeth that are disposed in front of the worm and engage with the outer circumferential surface saw teeth of the locking screw may be formed on the sidewall in a longitudinal direction.

[0025] According to the second embodiment of the present invention, the spacing-adjustable bone orthopedic apparatus may further include a second locking screw disposed in the through portion to be spaced apart from the locking screw, wherein the second locking screw may engage with the plate saw teeth and move in a longitudinal direction of the through portion.

Advantageous Effects

[0026] According to the present invention, the relative positions of two bone fragments can be easily finely adjusted after surgery by using a locking screw of which a locking state or a locking release state can be selectively set and which can perform both a function of moving a bone fragment and a function of fixing a plate to the bone fragment.

[0027] In addition, the relative positions of two bone fragments can be finely adjusted after surgery by rotating a locking screw using a worm.

DESCRIPTION OF DRAWINGS

[0028] FIG. 1 is a perspective view illustrating a conventional fixation plate for mandibular surgery.

[0029] FIG. 2 is a perspective view illustrating a state in which the fixation plate for mandibular surgery of FIG. 1 is attached to the mandible.

[0030] FIG. 3 is a perspective view illustrating a locking screw mounted on a plate in a spacing-adjustable bone orthopedic apparatus according to a first embodiment of the present invention.

[0031] FIG. 4 is an exploded view illustrating the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0032] FIG. 5 is a perspective view illustrating a locking state in which a screw body is engaged with a screw head in the locking screw of FIG. 3.

[0033] FIG. 6 is a perspective view illustrating a locking release state in which a state in which the screw body is engaged with the screw head in FIG. 3 is released.

[0034] FIG. 7 is a set of perspective views illustrating an operating process of the spacing-adjustable bone orthopedic apparatus according to the first embodiment of the present invention.

[0035] FIG. 8 is a perspective view illustrating an example in which an elastic member for assisting an upward force of the screw head is additionally mounted on the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0036] FIG. 9 is a perspective view illustrating an application example in which a ratchet method is applied to the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0037] FIG. 10 is a plan view illustrating an example in which the spacing-adjustable bone orthopedic apparatus of FIG. 3 includes two locking screws.

[0038] FIG. 11 is a plan view illustrating an application example in which a separate screw fixing hole is formed in the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0039] FIG. 12 is a perspective view illustrating a locking screw engaged with a worm in a spacing-adjustable bone orthopedic apparatus according to a second embodiment of the present invention.

[0040] FIG. 13 is an exploded view illustrating the spacing-adjustable bone orthopedic apparatus of FIG. 12.

[0041] FIG. 14 is a perspective view illustrating a second sidewall provided in the spacing-adjustable bone orthopedic apparatus according to the second embodiment of the present invention.

[0042] FIG. 15 is a plan view illustrating an application example in which the spacing-adjustable bone orthopedic apparatus of FIG. 14 includes a locking screw and a second locking screw.

MODES OF THE INVENTION

[0043] Hereinafter, specific content for implementing the present invention will be described with reference to the accompanying drawings. In addition, in description of the present invention, when it is determined that detailed description of related well-known functions which are obvious to those skilled in the art may unnecessarily obscure the gist of the present invention, such detailed description will be omitted.

[0044] A first embodiment of a spacing-adjustable bone orthopedic apparatus according to the present invention will be described below.

[0045] FIG. 3 is a perspective view illustrating a locking screw mounted on a plate in a spacing-adjustable bone orthopedic apparatus according to a first embodiment of the present invention. FIG. 4 is an exploded view illustrating the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0046] Referring to FIGS. 3 and 4, the spacing-adjustable bone orthopedic apparatus according to the present invention includes a plate 100 and a locking screw 200a.

[0047] The plate 100 includes a main body 110, a through portion 120 having a predetermined length and formed to pass through a central portion of the main body 110, and a sidewall 130a formed at one side of the main body 110. A plurality of plate saw teeth 150 extending toward the through portion 120 in a longitudinal direction of the sidewall 130a are formed on the sidewall 130a.

[0048] The plate 100 is disposed across a first bone and a second bone, which are two adjacent bone fragments, one side surface of the main body 110 of the plate 100 is supported by an upper surface of the first bone, and the other side surface thereof is supported by an upper surface of the second bone. Since the first bone and the second bone are spaced apart from each other, a certain region of the central portion of the main body 110 is a region which is not in contact with either the first bone or the second bone.

[0049] The through portion 120 has an elliptical shape formed to extend in a longitudinal direction of the main body 110. The through portion 120 is spaced a predetermined distance from the sidewall 130a and oriented parallel to the sidewall 130a. One side of the through portion 120 is disposed on the first bone, and the other side thereof is disposed on the second bone. A width of the through portion 120 is greater than a diameter of a body portion 211 of a screw body 210 and smaller than a diameter of a head portion 212 of the screw body 210.

[0050] The plurality of plate saw teeth 150 are formed on the sidewall 130a, and a second sidewall 140a faces the sidewall 130a and is formed at the other side of the plate 100. The sidewall 130a and the second sidewall 140a are formed to extend upward from one end portion and the other end portion of the plate, respectively. A distance between the second sidewall 140a and a longitudinal axis of the through portion 120 is greater than a distance between the sidewall 130a and the longitudinal axis of the through portion 120. The distance between the sidewall 130a and the longitudinal axis of the through portion 120 is approximately a radius of the screw head 220 of the locking screw 200a. A height of the sidewall 130a and a height of the second sidewall 140a may be the same as a height of the screw head 220.

[0051] The plate saw teeth 150 have a length greater than a length of the through portion 120 and are formed to extend in a longitudinal direction of an inner side surface of the sidewall 130a. A height of the extending plate saw teeth 150 may be the same as a height of extending outer circumferential surface saw teeth 222 formed on an outer circumferential surface of the screw head 220.

[0052] The locking screw 200a is inserted into the through portion 120. The locking screw 200a includes the screw body 210 and the screw head 220 which are capable of being coupled to each other. A hollow which is greater than the diameter of the head portion 212 of the screw body 210 is formed in the screw head 220 so that the screw body 210 may be inserted into the screw head 220. In a state in which the screw head 220 is disposed on the through portion 120 and fixed to the main body 110 of the plate 100, the screw

body 210 may be inserted into the first or second bone through the through portion 120 and coupled to an inner portion of the hollow of the screw head 220.

[0053] The screw body 210 includes the body portion 211 in which a screw thread 211a is formed on a lower outer circumferential surface of the screw body 210 and the head portion 212 in which locking saw teeth 212a are formed on an upper outer circumferential surface of the screw body 210. While the body portion 211 of the screw body 210 passes through the through portion 120 and is inserted into the first or second bone, a screw groove corresponding to the screw thread 211a is formed in the first or second bone by the screw thread 211a and engages with the screw thread 211a formed on the body portion 211. An insertion groove 212b into which a driver or the like is inserted is formed in an upper surface of the head portion 212 of the screw body 210. A user may insert the driver or the like into the insertion groove 212b and rotate the driver to insert the screw body 210 into the first or second bone.

[0054] The screw head 220 is disposed on the through portion 120 and supported by an upper surface of the main body 110 of the plate 100. The screw head 220 may be disposed between the sidewall 130a and the second sidewall 140a and may have the same height as the height of each of the sidewall 130a and the second sidewall 140a.

[0055] A locking protrusion 221 protruding inward is formed on a portion adjacent to a lowermost side of an inner circumferential surface of the screw head 220. While the screw body 210 is inserted into the first or second bone, a lower side surface of the head portion 212 of the screw body 210 is caught by the locking protrusion 221 of the screw head 220. The locking protrusion 221 extends to a length such that the head portion 212 of the screw body 210 is supported and the locking protrusion 221 does not come into contact with the body portion 211 of the screw body 210. The head portion 212 of the screw body 210 is caught by the locking protrusion 221 and is not inserted further downward.

[0056] The outer circumferential surface saw teeth 222 corresponding to a shape of the plate saw teeth 150 are formed on the outer circumferential surface of the screw head 220. The outer circumferential surface saw teeth 222 are formed along a circumference of a circumferential surface of the screw head 220. The outer circumferential surface saw teeth 222 may be formed to move in the longitudinal direction of the through portion 120 according to a set gear ratio when the screw head 220 rotates while engaging with the plate saw teeth 150.

[0057] FIG. 5 is a perspective view illustrating a locking state in which the screw body is engaged with the screw head in the locking screw of FIG. 3. FIG. 6 is a perspective view illustrating a locking release state in which a state in which the screw body is engaged with the screw head in FIG. 3 is released.

[0058] Referring to FIGS. 5 and 6, inner circumferential surface saw teeth 223 corresponding to a shape of the locking saw teeth 212a of the head portion 212 of the screw body 210 are formed on a part of the inner circumferential surface of the screw head 220. The inner circumferential surface saw teeth 223 are formed along a circumference of a part of the inner circumferential surface of the screw head 220. The inner circumferential surface saw teeth 223 may be formed on a lower region of the inner circumferential surface of the screw head. A height of the extending inner circumferential surface saw teeth 223 may be the same as a

height of the extending locking saw teeth **212a** of the head portion **212** of the screw body **210**. The inner circumferential surface saw teeth **223** are formed to engage with the locking saw teeth **212a** of the head portion **212** of the screw body **210** while the screw head **220** moves upward or downward with respect to the screw body **210**.

[0059] In the inner circumferential surface of the screw head **220**, a region in which the inner circumferential surface saw teeth **223** are formed is referred to as a locking region **s1**, and a region other than the locking region **s1** is referred to as a locking release region **s2**. The inner circumferential surface saw teeth **223** are formed in only the locking region **s1** of a part of the inner circumferential surface of the screw head **220**. Accordingly, since a height of the screw body **210** is changed in the hollow of the screw head **220**, a locking state in which the inner circumferential surface saw teeth **223** and the locking saw teeth **212a** are engaged and a locking release state in which the inner circumferential surface saw teeth **223** and the locking saw teeth **212a** are not engaged may be selectively formed. Meanwhile, a plurality of rotation grooves **225** into which an insertion tool (not shown) is inserted are formed in the inner circumferential surface of the screw head **220** in the locking release region **s2** in a longitudinal direction. The screw head **220** may be moved downward by inserting the insertion tool into the rotation groove **225** and pressing the insertion tool. In addition, the screw head **220** may be rotated by inserting the insertion tool into the rotation groove **225** and rotating the insertion tool.

[0060] When the inner circumferential surface saw teeth **223** are formed on a lower portion of the inner circumferential surface of the screw head **220**, while the screw head **220** moves upward, the locking state in which the locking saw teeth **212a** of the screw body **210** are engaged with the inner circumferential surface saw teeth **223** of the screw head **220** is formed. Meanwhile, while the screw head **220** moves downward, the locking release state in which a state in which the locking saw teeth **212a** of the screw body **210** are engaged with the inner circumferential surface saw teeth **223** of the screw head **220** is released is formed.

[0061] FIG. 7 is a set of perspective views illustrating an operating process of the spacing-adjustable bone orthopedic apparatus according to the first embodiment of the present invention.

[0062] Referring to FIG. 7, a surgical procedure using the spacing-adjustable bone orthopedic apparatus according to the present invention includes a preparation operation, a screw body insertion operation, a screw head push operation, a screw head rotation operation, and a screw head raising operation. Vertical movement or rotational movement is selectively applied to the screw head **220**. In this case, the operator may directly push or rotate the screw head **220** using a tool. Alternatively, vertical movement or rotational movement of the screw head **220** may be controlled using a driving part including a motor.

[0063] In the preparation operation, the operator arranges the plate **100** on both the first bone and the second bone so that one side of the plate **100** is in contact with a surface of the first bone and the other side of the plate **100** is in contact with the upper surface of the second bone. In this case, the screw head **220** is mounted on the plate **100**.

[0064] In the screw body insertion operation, the operator inserts the screw body **210** through the hollow of the screw head **220** and the through portion **120** of the plate **100** and

inserts and fixes the body portion **211** of the screw body **210** into the first or second bone. In this case, the head portion **212** of the screw body is caught and fixed on the locking protrusion **221**.

[0065] In a case in which the inner circumferential surface saw teeth **223** are formed on a lower portion of the inner circumferential surface of the screw head **220**, while the screw body **210** moves downward in the hollow of the screw head **220**, the locking state in which the locking saw teeth **212a** of the screw body **210** are engaged with the inner circumferential surface saw teeth **223** of the screw head **220** is formed (see FIG. 5).

[0066] In the screw head push operation, when the screw head **220** moves downward, the head portion **212** of the screw body **210** is positioned in an upper portion of the hollow of the screw head **220**. Accordingly, the locking release state in which engagement between the locking saw teeth **212a** of the screw body **210** and the inner circumferential surface saw teeth **223** of the screw head **220** is released is formed (see FIG. 6).

[0067] In the screw head rotation operation, the screw head **220** is rotated in one or the other direction to move to one side or the other side in the longitudinal direction of the through portion **120**. Although the engagement between the screw body **210** inserted into the hollow of the screw head **220** and the screw head **220** is released, the screw body **210** moves with the movement of the screw head **220**. A distance between the first bone and the second bone increases or decreases according to a rotational direction of the screw head **220** to adjust a distance between the first bone and the second bone.

[0068] In the screw head raising operation, the screw head **220** may naturally move upward due to the inherent elasticity of bone. Accordingly, when the locking saw teeth **212a** of the screw body **210** engage with the inner circumferential surface saw teeth **223** of the screw head **220**, the locking release state is switched to the locking state. In the locking state, since the screw body **210** is fixed to the first bone or second bone, the screw head **220** engaged with the screw body **210** does not easily move along the plate saw teeth **150** of the plate **100**. Due to the inherent elasticity of bone, the screw head **220** may move upward about 0.5 mm to 3 mm.

[0069] FIG. 8 is a perspective view illustrating an example in which an elastic member for assisting an upward force of the screw head is additionally mounted on the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0070] Referring to FIG. 8, the spacing-adjustable bone orthopedic apparatus according to the embodiment of the present invention may include an elastic member **400** which is disposed between a lower surface of the screw head **220** and the main body **110** and of which one side is inclined toward the lower surface of the screw head **220**. The elastic member **400** may be a washer, that is, a spring washer, which extends obliquely from one end portion to the other end portion and in which a height of one end portion and a height of the other end portion are different. Accordingly, the elastic member **400** may provide an additional upward force in addition to the inherent elasticity of bone.

[0071] FIG. 9 is a perspective view illustrating an application example in which a ratchet method is applied to the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0072] Referring to FIG. 9, the spacing-adjustable bone orthopedic apparatus according to the first embodiment of the present invention may include head ratchet saw teeth **224**

formed on the lower side surface of the screw head 220 and body ratchet saw teeth 111 which are formed on the upper side surface of the main body 110 and correspond to a shape of the head ratchet saw teeth 224. When the screw head 220 rotates, the head ratchet saw teeth 224 of the screw head 220 engage with the body ratchet saw teeth 111 of the main body 110. Accordingly, whenever the screw head 220 rotates, a click sound is generated, a frictional force may be provided to the screw head 220 to precisely control the rotation of the screw head 220, and the screw head 220 may also be prevented from idling on the main body 110 when stopped.

[0073] FIG. 10 is a plan view illustrating an example in which the spacing-adjustable bone orthopedic apparatus of FIG. 3 includes two locking screws.

[0074] Referring to FIG. 10, the spacing-adjustable bone orthopedic apparatus according to the embodiment of the present invention may include a first locking screw 200a and a second locking screw 200b. The second locking screw 200b is disposed in the through portion 120 to be spaced apart from the first locking screw 200a in the longitudinal direction of the through portion 120. The first locking screw 200a is disposed on the upper side of the first bone, and the second locking screw 200b is disposed on the upper side of the second bone. Accordingly, in a state in which a position of the second locking screw 200b is fixed with respect to the second bone, the first locking screw 200a may be operated to adjust the movement of the first bone, and in a state in which a position of the first locking screw 200a is fixed with respect to the first bone, the second locking screw 200b may be operated to adjust the movement of the second bone.

[0075] FIG. 11 is a plan view illustrating an application example in which a separate screw fixing hole is formed in the spacing-adjustable bone orthopedic apparatus of FIG. 3.

[0076] Referring to FIG. 11, in the plate 100 of the spacing-adjustable bone orthopedic apparatus according to the embodiment of the present invention, a screw fixing hole 160 spaced apart from the through portion 120 may be formed. In a state in which the plate 100 is fixed to the first or second bone using the screw fixing hole 160 and a fixing screw 161, the locking screw 200a may be operated. Accordingly, a fixing force of the plate 100 increases so that a more stable surgical procedure may be performed.

[0077] A second embodiment of a spacing-adjustable bone orthopedic apparatus according to the present invention will be described below.

[0078] FIG. 12 is a perspective view illustrating a locking screw engaged with a worm in the spacing-adjustable bone orthopedic apparatus according to the second embodiment of the present invention. FIG. 13 is an exploded view illustrating the spacing-adjustable bone orthopedic apparatus of FIG. 12.

[0079] Referring to FIGS. 12 and 13, the spacing-adjustable bone orthopedic apparatus according to the second embodiment of the present invention includes a plate 100, a locking screw 200b, and a worm 300.

[0080] The plate 100 includes a main body 110, a through portion 120 having a predetermined length and passing through the main body 110, and a first support part 131 and a second support part 132 which are formed to protrude from a bottom surface at one side of the main body 110 and are spaced apart from each other. The plate 100 are disposed across a first bone and a second bone, which are two adjacent bone fragments, one side surface of the main body 110 of the

plate 100 is supported by an upper surface of the first bone, and the other side surface thereof is supported by an upper surface of the second bone.

[0081] A sidewall 130b is formed at one side of the main body 110 in a longitudinal direction of the main body 110 and has a predetermined height. Each of the first support part 131 and the second support part 132 is formed in the form of a wall extending inward from the sidewall 130b. The worm 300 is disposed between the sidewall 130b and the through portion 120. A distance between the first support part 131 and the second support part 132 is greater than a length of the through portion 120. One end portion of the worm 300 passes through and is supported by the first support part 131, and the other end portion of the worm 300 passes through and is supported by the second support part 132.

[0082] The through portion 120 has an elliptical shape formed to extend in the longitudinal direction of the main body 110. One side of the through portion 120 may be disposed on the first bone, and the other side thereof is disposed on the second bone. Alternatively, when a hole is formed outside the through portion 120 to fasten a screw, the through portion 120 may be disposed on only one bone of the first bone and the second bone. A width of the through portion 120 is greater than a diameter of a body portion 211 of a screw body 210 and smaller than a diameter of a head portion 212 of the screw body 210.

[0083] The locking screw 200b is inserted into the through portion 120. The locking screw 200b includes the screw body 210 and the screw head 220 which are capable of being coupled to each other. Outer circumferential surface saw teeth 222 are formed on an outer circumferential surface of the screw head 220. The outer circumferential surface saw teeth 222 are formed along a circumference of the outer circumferential surface of the screw head 220. A hollow having a diameter greater than or equal to the diameter of the head portion 212 of the screw body 210 may be formed in the screw head 220, and the screw body 210 may be inserted into the hollow. In a state in which the screw head 220 is disposed on the through portion 120 and the main body 110 of the plate 100, the screw body 210 passes through the through portion 120 so that an end portion of the screw body 210 is inserted into the first or second bone.

[0084] A locking protrusion 221 protruding inward is formed on a portion adjacent to a lowermost side of an inner circumferential surface of the screw head 220. While the screw body 210 is inserted into the first or second bone, a lower side surface of the head portion 212 of the screw body 210 is caught by the locking protrusion 221 of the screw head 220. The locking protrusion 221 extends inward to a length such that the head portion 212 of the screw body 210 is supported and the locking protrusion 221 does not come into contact with an outer circumferential surface of the body portion 211 of the screw body 210. The head portion 212 of the screw body 210 is caught by the locking protrusion 221 and is not inserted further downward.

[0085] The worm 300 is supported by the first support part 131 and the second support part 132, and a screw thread that engages with the outer circumferential surface saw teeth 222 of the locking screw 200b is formed on the worm 300, wherein the locking screw 200b is a worm wheel. The worm 300 is disposed between the sidewall 130b and the locking screw 200b. The screw thread corresponds to the through portion 120 and is formed in a longitudinal direction of the

worm **300** and has a length approximately as long as the through portion **120** extends. The outer circumferential surface saw teeth **222** of the locking screw **200b** are formed so that the locking screw **200b** moves according to a preset gear ratio in a longitudinal direction of the through portion **120** when the screw head **200** rotates while engaging with the screw thread of the worm **300**. After the locking screw **200b** moves a predetermined distance, reverse rotation of the screw head **200** of the locking screw **200b** is prevented by the worm **300**, and the screw head **200** is stopped at the position to which it moves.

[0086] A motor (not shown) may be connected to one side or the other side of the worm **300**. The motor may rotate the worm **300** in one or the other direction. In addition, the motor may be mounted on the plate **100** and designed to be integrated with the plate **100**.

[0087] Meanwhile, a screw fixing hole **160** spaced apart from the through portion **120** may be formed in the plate **100**. In a state in which the plate **100** is fixed to the first or second bone using the screw fixing hole **160** and a fixing screw **161**, the locking screw **200b** may be operated. Accordingly, a fixing force of the plate **100** can increase, and a more stable surgical procedure can be performed.

[0088] FIG. **14** is a perspective view illustrating a second sidewall provided in the spacing-adjustable bone orthopedic apparatus according to the second embodiment of the present invention.

[0089] Referring to FIG. **14**, the spacing-adjustable bone orthopedic apparatus of the present invention may further include a second sidewall **140b**. A plurality of plate saw teeth **150** that are disposed in front of the worm **300** and engage with the outer circumferential surface saw teeth **222** of the locking screw **200b** are formed on the second sidewall **140b** in a longitudinal direction. The second sidewall **140b** faces the sidewall **130b** and is formed at the other side of the plate **100**. A distance between the sidewall **130b** and the through portion **120** is greater than a distance between the second sidewall **140b** and the through portion **120**. A distance between the second sidewall **140b** and a longitudinal axis of the through portion **120** may be approximately a radius of the screw head **220** of the locking screw **200b**. A height of the sidewall **130b** and a height of the second sidewall **140b** may be approximately a height of the screw head **220**.

[0090] The plate saw teeth **150** have a length greater than a length of the through portion **120** and are formed to extend in the longitudinal direction of an inner side surface of the second sidewall **140b**. The height of the plate saw teeth **150** may be almost the same as a height of the outer circumferential surface saw teeth **222** formed on the outer circumferential surface of the screw head **220**.

[0091] FIG. **15** is a plan view illustrating an application example in which the spacing-adjustable bone orthopedic apparatus of FIG. **14** includes the locking screw and a second locking screw.

[0092] Referring to FIG. **15**, the spacing-adjustable bone orthopedic apparatus according to the second embodiment of the present invention may include the locking screw **200b** and a second locking screw **500**. The second locking screw **500** is disposed in the through portion **120** to be spaced apart from the locking screw **200b**. The structure of the locking screw which is used in the spacing-adjustable bone orthopedic apparatus according to the first embodiment of FIG. **5** and in which locking or locking release is possible may be

applied to the second locking screw **500** without change. The locking screw **200b** is disposed on the first bone, and the second locking screw **500** is disposed on the second bone. [0093] It is designed that the worm **300** engages with the locking screw **200b**, and the worm **300** does not engage with the second locking screw **500**. In addition, both the locking screw **200b** and the second locking screw **500** are designed to engage with the plate saw teeth **150**. In a state in which a position of the second locking screw **500** is fixed with respect to the second bone, when the worm **300** is rotated, the locking screw **200b** may be moved toward one side or the other side in the longitudinal direction of the through portion **120** to adjust the movement of the first bone. Meanwhile, in a state in which a position of the locking screw **200b** is fixed with respect to the first bone, an operator may move the second locking screw **500** toward one side or the other side in the longitudinal direction of the through portion **120** to adjust the movement of the second bone. Specifically, the operator may rotate a screw head **520** of the second locking screw **500** using a tool or the like after fixing a screw body **510** of the second locking screw **500** to the second bone. Accordingly, outer circumferential surface saw teeth **521** provided on the screw head **520** of the second locking screw **500** engage with the plate saw teeth **150**, and the second locking screw **500** is moved.

[0094] The scope of the present invention in the art is not limited to the content and description of the embodiment described above. In addition, it is mentioned once again that the scope of the present invention is not limited by obvious changes or substitutions in the art to which the present invention belongs.

REFERENCE NUMERALS

[0095]	100: PLATE
[0096]	110: MAIN BODY
[0097]	111: BODY RACHET SAW TEETH
[0098]	120: THROUGH PORTION
[0099]	130A, B: SIDEWALL
[0100]	131: FIRST SUPPORT PART
[0101]	132: SECOND SUPPORT PART
[0102]	140A, B: SECOND SIDEWALL
[0103]	150: PLATE SAW TEETH
[0104]	160: SCREW FIXING HOLE
[0105]	161: FIXING SCREW
[0106]	200: LOCKING SCREW
[0107]	200A, B: LOCKING SCREW
[0108]	210: SCREW BODY
[0109]	211: BODY PORTION
[0110]	211A: BODY PORTION SCREW THREAD
[0111]	212: HEAD PORTION
[0112]	212A: LOCKING SAW TEETH
[0113]	212B: INSERTION GROOVE
[0114]	220: SCREW HEAD
[0115]	221: LOCKING PROTRUSION
[0116]	222: OUTER CIRCUMFERENTIAL SURFACE SAW TEETH
[0117]	223: INNER CIRCUMFERENTIAL SURFACE SAW TEETH
[0118]	224: HEAD RACHET SAW TEETH
[0119]	225: ROTATION GROOVE
[0120]	300: WORM
[0121]	400: ELASTIC MEMBER
[0122]	500: SECOND LOCKING SCREW

[0123] s1: LOCKING REGION

[0124] s2: LOCKING RELEASE REGION

1. A spacing-adjustable bone orthopedic apparatus comprising:

a plate including a main body and a through portion having a predetermined length and formed to pass through the main body; and

a locking screw which is inserted into the through portion, which includes a screw body and a screw head which are coupled to each other, and in which outer circumferential surface saw teeth are formed on an outer circumferential surface of the screw head.

2. The spacing-adjustable bone orthopedic apparatus of claim 1, wherein:

the plate includes a sidewall formed at one side of the main body;

a plurality of plate saw teeth are formed on the sidewall in a longitudinal direction of the sidewall to extend toward the through portion;

the screw body includes a body portion in which a screw thread is formed on a lower outer circumferential surface of the screw body and a head portion in which locking saw teeth are formed on an upper outer circumferential surface of the screw body;

the outer circumferential surface saw teeth of the screw head correspond to a shape of the plate saw teeth; and inner circumferential surface saw teeth corresponding to a shape of the locking saw teeth of the head portion of the screw body are formed on a part of an inner circumferential surface of the screw head.

3. The spacing-adjustable bone orthopedic apparatus of claim 2, wherein the screw head is disposed on the through portion and supported by an upper surface of the main body of the plate.

4. The spacing-adjustable bone orthopedic apparatus of claim 2, wherein the inner circumferential surface of the screw head includes:

a locking region in which the inner circumferential surface saw teeth are formed; and

a locking release region other than the locking region, wherein the locking region is formed to extend inside the inner circumferential surface of the screw head and has a predetermined height.

5. The spacing-adjustable bone orthopedic apparatus of claim 2, wherein:

a locking protrusion protruding inward is formed on a portion adjacent to a lowermost side of the inner circumferential surface of the screw head; and

a lower side surface of the head portion of the screw body is caught by the locking protrusion.

6. The spacing-adjustable bone orthopedic apparatus of claim 2, further comprising an elastic member which is disposed between a lower surface of the screw head and the main body and of which one side is inclined toward the lower surface of the screw head.

7. The spacing-adjustable bone orthopedic apparatus of claim 6, wherein the elastic member includes a spring washer.

8. The spacing-adjustable bone orthopedic apparatus of claim 2, further comprising:

head ratchet saw teeth formed on a lower side surface of the screw head; and

body ratchet saw teeth formed on an upper side surface of the main body and corresponding to a shape of the head ratchet saw teeth.

9. The spacing-adjustable bone orthopedic apparatus of claim 2, wherein the locking screw includes:

a first locking screw; and

a second locking screw disposed in the through portion to be spaced apart from the first locking screw in a longitudinal direction of the through portion.

10. The spacing-adjustable bone orthopedic apparatus of claim 2, wherein a screw fixing hole spaced apart from the through portion is formed in the plate.

11. The spacing-adjustable bone orthopedic apparatus of claim 1, wherein the plate includes a first support part and a second support part which are formed to protrude from a bottom surface at one side of the main body and are spaced apart from each other,

wherein the spacing-adjustable bone orthopedic apparatus includes a worm which is supported by the first support part and the second support part and on which a screw thread that engages with the outer circumferential surface saw teeth of the locking screw is formed, and the locking screw is used as a worm wheel.

12. The spacing-adjustable bone orthopedic apparatus of claim 11, wherein:

one end portion of the worm passes through and is supported by the first support part; and

the other end portion of the worm passes through and is supported by the second support part.

13. The spacing-adjustable bone orthopedic apparatus of claim 11, further comprising a motor which is connected to the worm and rotates the worm in one direction or the other direction.

14. The spacing-adjustable bone orthopedic apparatus of claim 11, wherein:

the plate further includes a sidewall formed at the other side of the main body; and

a plurality of plate saw teeth that are disposed in front of the worm and engage with the outer circumferential surface saw teeth of the locking screw are formed on the sidewall in a longitudinal direction.

15. The spacing-adjustable bone orthopedic apparatus of claim 14, further comprising a second locking screw disposed in the through portion to be spaced apart from the locking screw,

wherein the second locking screw engages with the plate saw teeth and moves in a longitudinal direction of the through portion.

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